

Node.js for PHP Developers - Study Notes (Volume 1)

Introduction

These notes are designed for developers migrating from PHP to Node.js. They focus on explaining Node.js concepts using familiar PHP examples.

Why Node.js?

JavaScript runs on the server using the V8 engine. Single-threaded, event-driven architecture. Excellent for APIs, real-time applications, and microservices. Huge npm ecosystem.

PHP vs Node.js

PHP Node.js echoconsole.log() \$_GETreq.query \$_POSTreq.body \$_FILESmulter mysql/ PDOmysql2 / Prisma / Sequelize include/requireimport / require header()res.status(), res.set() json_encode()res.json()

JavaScript Essentials

Learn: Variables (let, const) Objects and Arrays Functions & Arrow Functions Destructuring Promises Async/Await

Node.js Core Concepts

Runtime & V8 Engine Event Loop Non-blocking I/O Modules (CommonJS & ESM) package.json & npm

Learning Roadmap

1. JavaScript Fundamentals
2. Node.js Core
3. Express.js
4. REST APIs
5. Databases (MySQL, PostgreSQL, MongoDB)
6. Authentication (JWT, bcrypt)
7. Validation & Error Handling
8. Security
9. Docker & Deployment
10. Build Production Projects

Recommended Projects

Todo API Authentication System HRMS Backend E-commerce Backend Chat Application with Socket.IO

Next Steps

Future volumes can cover Express.js, Prisma, Authentication, Docker, Testing, System Design, and a complete PHP-to-Node migration guide with production examples.